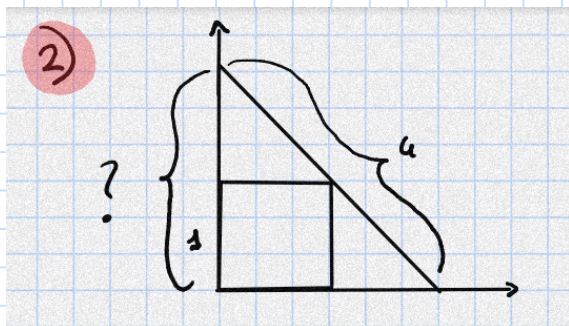
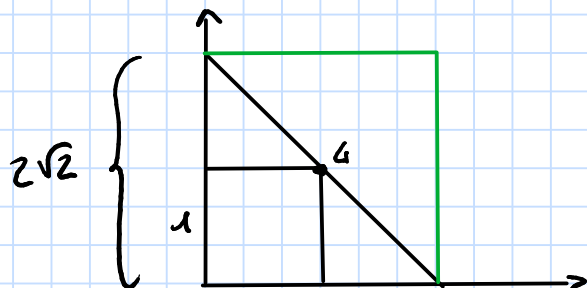


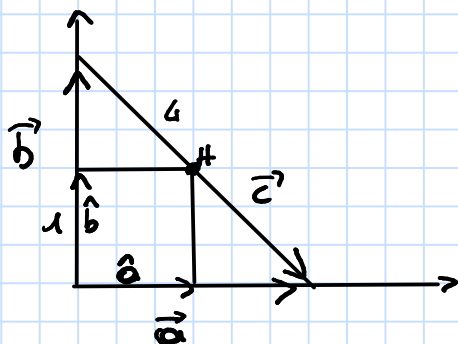


Diagonale quadrato = $l\sqrt{2}$



$$4 = l \cdot \sqrt{2}$$

$$l = \frac{4}{\sqrt{2}} \cdot \frac{\sqrt{2}}{\sqrt{2}} = \frac{4\sqrt{2}}{2} = 2\sqrt{2}$$



$$\vec{c} = \vec{a} - \vec{b}$$

$$c = 4$$

$$c^2 = \vec{c} \cdot \vec{c} = 16$$

$$c^2 = (\vec{a} - \vec{b}) \cdot (\vec{a} - \vec{b}) = 16$$

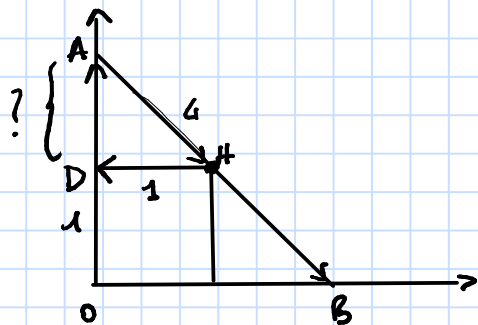
$$\vec{a} = a \cdot \hat{a}$$

$$\vec{b} = b \cdot \hat{b}$$

$$c^2 = (a \cdot \hat{a} - b \cdot \hat{b}) \cdot (a \cdot \hat{a} - b \cdot \hat{b}) = 16$$

$$c^2 = a^2 \hat{a}^2 - ab \hat{a} \hat{b} - ab \hat{a} \hat{b} + b^2 \hat{b}^2 = 16$$

$$c^2 = a^2 + b^2 = 16 \quad \text{triangolo rettangolo}$$



$$H = \lambda B + (1-\lambda)A \quad \text{con } \overline{AB} = 4$$

$$\overrightarrow{AH} = H - A$$

$$\overrightarrow{DH} = 1 = D - H$$

$$\overrightarrow{AD} = \overrightarrow{AH} + \overrightarrow{DH}$$

$$AO^2 = \overrightarrow{AD} \cdot \overrightarrow{AD}$$

$$AD^2 + DH^2 = AH^2 \quad \text{pitagora}$$

$$AH^2 = \overrightarrow{AH} \cdot \overrightarrow{AH} = [(\overrightarrow{AH} + \overrightarrow{DH}) + (D-H)] \cdot [(\overrightarrow{AH} + \overrightarrow{DH}) + (D-H)] =$$

$$= \left\{ [(\overrightarrow{AH} + \overrightarrow{DH}) \cdot (\overrightarrow{AH} + \overrightarrow{DH})] + (D-H)^2 \right\} \cdot \left\{ [(\overrightarrow{AH} + \overrightarrow{DH}) \cdot (\overrightarrow{AH} + \overrightarrow{DH})] + (D-H)^2 \right\} =$$

$$= \left\{ [\overrightarrow{AH} \cdot \overrightarrow{AH} + \overrightarrow{AH} \cdot \overrightarrow{DH} + \overrightarrow{DH} \cdot \overrightarrow{AH} + \overrightarrow{DH} \cdot \overrightarrow{DH}] + D^2 + H^2 - 2DH \right\} \cdot \left\{ [\overrightarrow{AH} \cdot \overrightarrow{AH} + \overrightarrow{AH} \cdot \overrightarrow{DH} + \overrightarrow{DH} \cdot \overrightarrow{AH} + \overrightarrow{DH} \cdot \overrightarrow{DH}] + D^2 + H^2 - 2DH \right\} =$$

$$= \left\{ AH^2 + DH^2 + 2\overrightarrow{AH} \cdot \overrightarrow{DH} + D^2 + H^2 - 2DH \right\} \cdot \left\{ AH^2 + DH^2 + 2\overrightarrow{AH} \cdot \overrightarrow{DH} + D^2 + H^2 - 2DH \right\} =$$

prodotto
vettoriale
tra due vettori
a un'angolo

$$= \left\{ AH^2 + DH^2 + D^2 + H^2 - 2DH \right\} \cdot \left\{ AH^2 + DH^2 + D^2 + H^2 - 2DH \right\} =$$

$$= \left\{ (H-A)^2 + (D-H)^2 + D^2 + H^2 - 2(D-H) \right\} \cdot \left\{ (H-A)^2 + (D-H)^2 + D^2 + H^2 - 2(D-H) \right\} =$$

$$= \left\{ H^2 + A^2 - 2AH + D^2 + H^2 - 2DH + D^2 + H^2 - 2D + 2H \right\} \cdot \left\{ H^2 + A^2 - 2AH + D^2 + H^2 - 2DH + D^2 + H^2 - 2D + 2H \right\} =$$

$$= \left\{ A^2 + 3H^2 + 2D^2 - 2H + 2A - 2D + 2H - 2D + 2H \right\} \cdot \left\{ A^2 + 3H^2 + 2D^2 - 2H + 2A - 2D + 2H - 2D + 2H \right\} =$$

$$= \left\{ A^2 + 3H^2 + 2D^2 + 2A - 4D + 2H \right\} \cdot \left\{ A^2 + 3H^2 + 2D^2 + 2A - 4D + 2H \right\} =$$

$$= A^2 \cdot A^2 + A^2 \cdot 3H^2 + A^2 \cdot 2D^2 + A^2 \cdot 2A + A^2 \cdot (-4D) + A^2 \cdot 2H +$$

$$+ 3H^2 \cdot A^2 + 3H^2 \cdot 3H^2 + 3H^2 \cdot 2D^2 + 3H^2 \cdot 2A + 3H^2 \cdot (-4D) + 3H^2 \cdot 2H +$$

$$+ 2D^2 \cdot A^2 + 2D^2 \cdot 3H^2 + 2D^2 \cdot 2D^2 + 2D^2 \cdot 2A + 2D^2 \cdot (-4D) + 2D^2 \cdot (2H) +$$

$$+ 2A \cdot A^2 + 2A \cdot 3H^2 + 2A \cdot 2D^2 + 2A \cdot 2A + 2A \cdot (-4D) + 2A \cdot 2H +$$

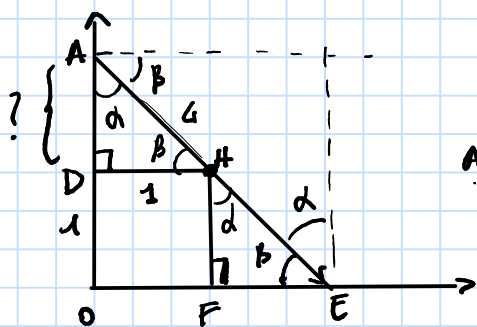
$$+ (-4D) \cdot A^2 + (-4D) \cdot 3H^2 + (-4D) \cdot 2D^2 + (-4D) \cdot 2A + (-4D) \cdot (-4D) + (-4D) \cdot 2H +$$

$$+ 2H \cdot A^2 + 2H \cdot 3H^2 + 2H \cdot 2D^2 + 2H \cdot 2A + 2H \cdot (-4D) + 2H \cdot 2H =$$

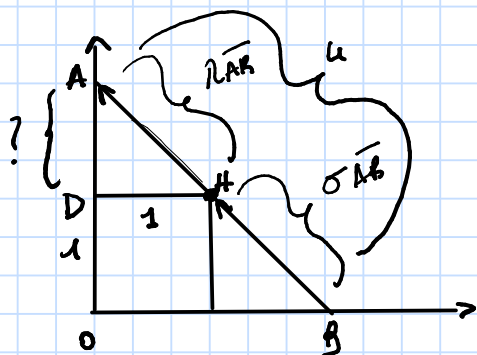
$$\begin{aligned}
&= A^2 \cdot A^2 + A^2 \cdot 3H^2 + A^2 \cdot 2D^2 + A^2 \cdot 2A + A^2 \cdot (-4D) + A^2 \cdot 2H + \\
&+ 3H^2 \cdot A^2 + 3H^2 \cdot 3H^2 + 3H^2 \cdot 2D^2 + 3H^2 \cdot 2A + 3H^2 \cdot (-4D) + 3H^2 \cdot 2H + \\
&+ 2D^2 \cdot A^2 + 2D^2 \cdot 3H^2 + 2D^2 \cdot 2D^2 + 2D^2 \cdot 2A + 2D^2 \cdot (-4D) + 2D^2 \cdot (2H) + \\
&+ 2A \cdot A^2 + 2A \cdot 3H^2 + 2A \cdot 2D^2 + 2A \cdot 2A + 2A \cdot (-4D) + 2A \cdot 2H + \\
&+ (-4D) \cdot A^2 + (-4D) \cdot 3H^2 + (-4D) \cdot 2D^2 + (-4D) \cdot 2A + (-4D) \cdot (-4D) + (-4D) \cdot 2H + \\
&+ 2H \cdot A^2 + 2H \cdot 3H^2 + 2H \cdot 2D^2 + 2H \cdot 2A + 2H \cdot (-4D) + 2H \cdot 2H =
\end{aligned}$$

$$\begin{aligned}
&= A^4 + 3A^2H^2 + 2A^2D^2 + 2A^3 - 4AD^2 + 2A^2H + 3A^2H^2 + 9H^2 + 6H^2D^2 + 6H^2A - 12H^2D + 6H^3 + \\
&+ 2A^2D^2 + 6D^2H^2 + 4D^4 + 4D^2A - 8D^3 + 4D^2H + 2A^3 + 6H^2A + 4D^2A + 4A^2 - 8AD + 4AH - 4A^2D + \\
&- 12H^2D - 8D^3 - 8DA + 16D^2 - 8DH + 2A^2H + 6H^3 + 4D^2H + 4AH - 8HD + 4H^2
\end{aligned}$$

$$\begin{aligned}
&= A^4 + 4D^4 + 4A^3 + 12H^3 - 16D^3 + 4A^2 + 13H^2 + 16D^2 + 6A^2H + 6AD^2 + 12D^2H^2 - 8A^2D + 4A^2H + 12H^2A - 26HD \\
&+ 8D^2A + 8D^2H - 16AD + 8AH - 16DH
\end{aligned}$$



$$\begin{aligned}
\frac{AH + HE}{\sin 90^\circ} &= \frac{FE + AH}{\sin \alpha} = \frac{HF + AH}{\sin \beta} \\
\frac{4}{\sin 90^\circ} &= \frac{FE + AH}{\sin \alpha} = \frac{HF + AH}{\sin \beta}
\end{aligned}$$

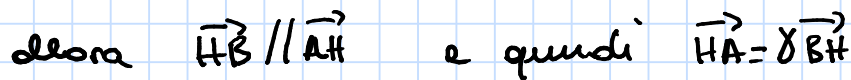
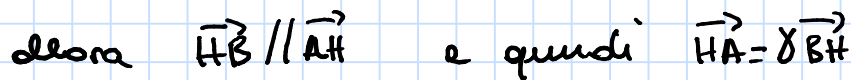


$$A = (0, 1) \quad B = (1, 0) \quad H = (1, 1)$$

$$AB = \lambda AH + \sigma HB = 4$$

$$\vec{AB} = \lambda \vec{AH} + \sigma \vec{HB} = 4$$

$$A - B = \lambda(A - H) + \sigma(H - B) = 4 \rightarrow \lambda A - \lambda H + \sigma H - \sigma B = A - B \rightarrow$$



ma le deriazioni sono tutte le stesse perché sono tutti paralleli tra loro

$$\frac{BA \cdot \hat{BA}}{\hat{BA}} = \alpha \frac{HA \cdot \hat{BA}}{\hat{BA}} = \frac{\gamma BA \cdot \hat{BA}}{\hat{BA}}$$

$$u = (0, 1+y) - (1+x, 0) = d(0, 1+y) - (1, 1) = \delta(1, 1) - (1+x, 0)$$

$$u = (-1-x, 1+y) = \alpha(-1, 1) = \delta(-x, y)$$

$$\begin{pmatrix} -1-x & 1-y \end{pmatrix} \begin{pmatrix} -1-x \\ 1-y \end{pmatrix} = x^2 + 2x + 1 + y^2 - 2y + 1$$

$$BA = \sqrt{\vec{BA} \cdot \vec{BA}} = \sqrt{x^2 + y^2 + 2x - 2y + 2} = 4$$

$$HA = \sqrt{\vec{HA} \cdot \vec{HA}} = \sqrt{y^2 + 1}$$

$$BH = \sqrt{\overrightarrow{BA} \cdot \overrightarrow{BH}} = \sqrt{x^2 + 1}$$

$$\begin{pmatrix} -1 & y \end{pmatrix} \begin{pmatrix} -1 \\ y \end{pmatrix} = 1 + y^2$$

$$)^{-x} \quad 1) \begin{pmatrix} -x & 1 \\ 1 & \end{pmatrix} = x^2 + 1$$

$$16 = 8(x^2 + 1) \rightarrow 8 = \frac{16}{x^2 + 1}$$

